SECURE BOOT构建

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一、编译生成签名bin

编译生成签名bin，同时打开软件secboot相关code。
验签功能是通过efuse来控制，详细见1.2

需要修改和增加文件如下：

banyan/gx306l_h3/board_info.json
banyan/gx306l_h3/fmd_partition_table.json
banyan/gx306l_h3/key/bl_priv_key.pem Added
banyan/gx306l_h3/key/bl_pub_key.pem Added
banyan/gx306l_h3/pkg/pkg_cfg.json

1.1 修改board info

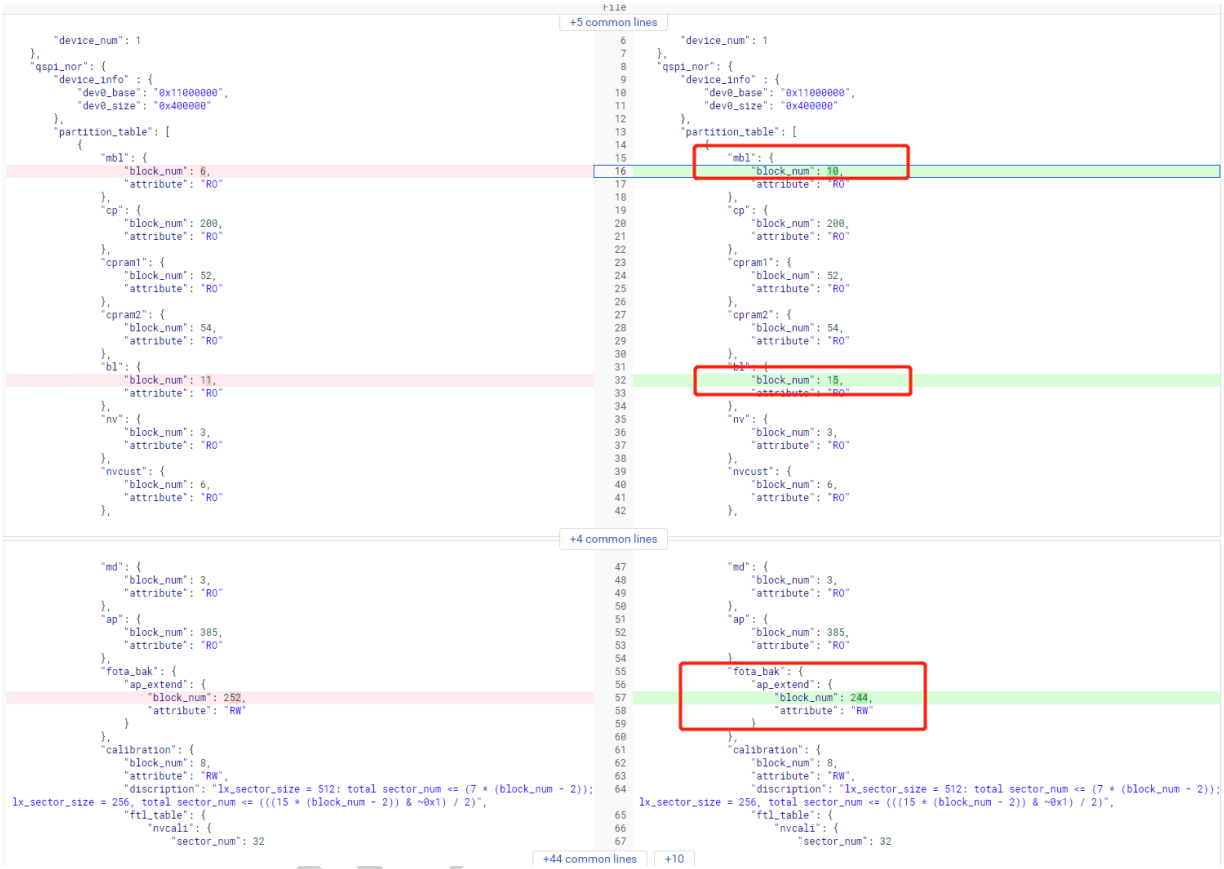
主要修改board_info.json.
Board info 没有配置security项或者指定value值是NSEC，那么当前board就不会开启编译签名。
开启secboot配置如下：



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1.2 分区调整：

主要修改发fmd_partition_table.jason. 打开secboot code size会增大16k



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1.3 添加验签的密钥对

产生密钥对需要使用到我们的python脚本，如下：

脚本位置：build/sign_script下



产生密钥对有两种方式：

1、采用归芯的脚本生成密钥对：

执行genRSAkey.py，生成的密钥对，这种密钥对我们是可以直接使用的。

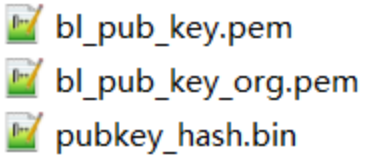
```
$ python genRSAkey.py bl
```

此脚本会产生需要使用的密钥对和pubkey的hash

2、使用openssl等其他工具产生密钥对，此密钥的公钥需要用convert_pubkey.py脚本转换才能被归芯使用。

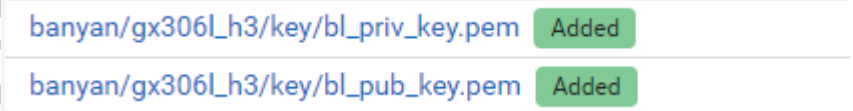
```
lanqing.liu@LAPTOP-6GMMSPHA MINGW64 /d/workspace/test/sign_script (master)
$ python convert_pubkey.py ./out/bl_pub_key_org.pem  ./out/bl_pub_key.pem ./out/pubkey_hash.bin
```

会生成如下文件：

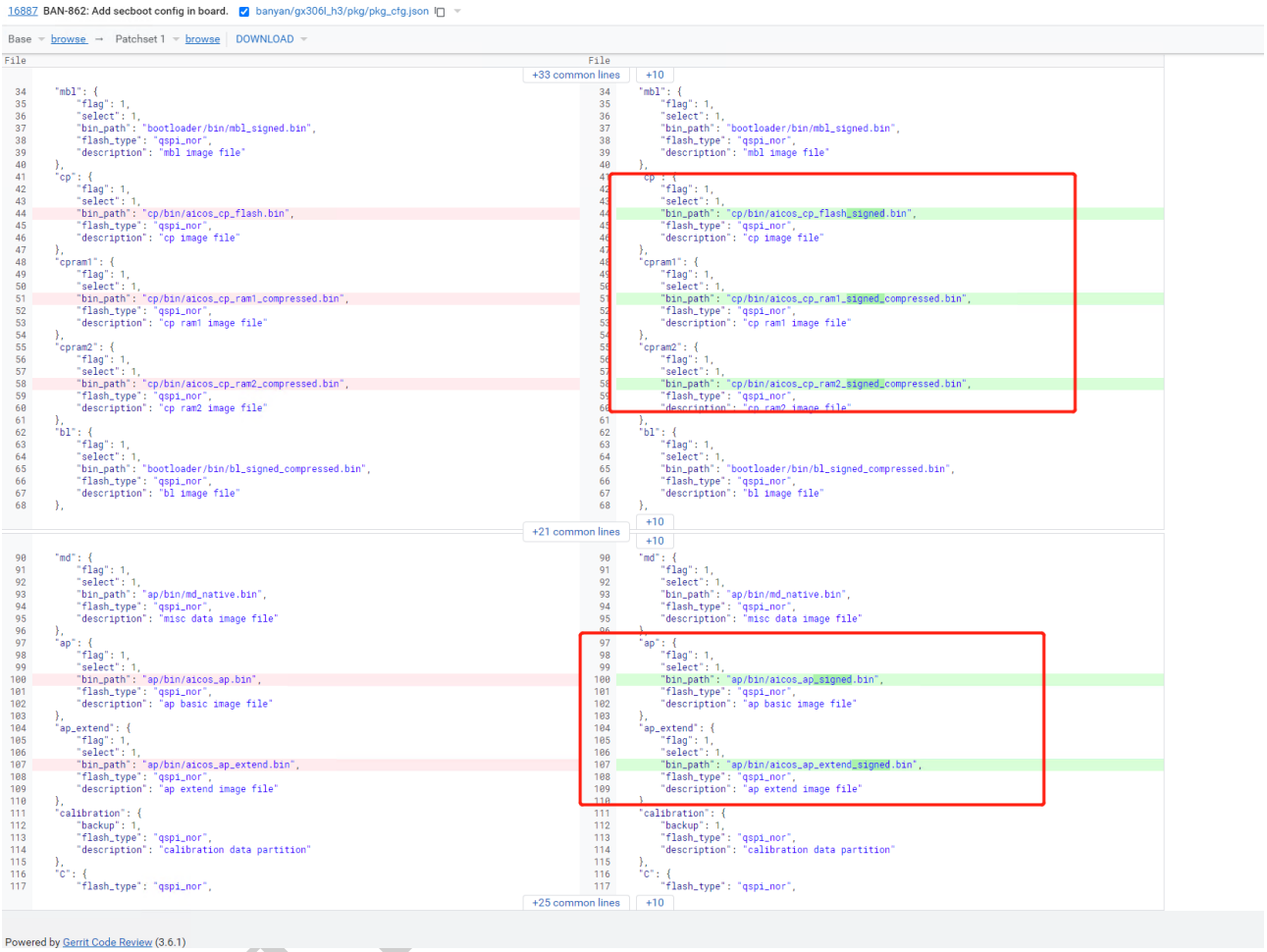


将此转换后的pubkey和之前生成的privkey放到如下目录：

例如：targets/banyan/gx306l_h3



1.4 修改pkg生成包含的签名bin



16887 BAN-862: Add secboot config in board. banyan/gx306l_h3/pkg/pkg_cfg.json

Base browse → Patchset 1 browse DOWNLOAD

File File

+33 common lines +10

```
"mbl": {
    "flag": 1,
    "select": 1,
    "bin_path": "bootloader/bin/mbl_signed.bin",
    "flash_type": "qspi_nor",
    "description": "mbl image file"
},
"cp": {
    "flag": 1,
    "select": 1,
    "bin_path": "cp/bin/aicos_cp_flash.bin",
    "flash_type": "qspi_nor",
    "description": "cp image file"
},
"cpram1": {
    "flag": 1,
    "select": 1,
    "bin_path": "cp/bin/aicos_cp_ram1_compressed.bin",
    "flash_type": "qspi_nor",
    "description": "cp ram1 image file"
},
"cpram2": {
    "flag": 1,
    "select": 1,
    "bin_path": "cp/bin/aicos_cp_ram2_compressed.bin",
    "flash_type": "qspi_nor",
    "description": "cp ram2 image file"
},
"bl": {
    "flag": 1,
    "select": 1,
    "bin_path": "bootloader/bin/bl_signed_compressed.bin",
    "flash_type": "qspi_nor",
    "description": "bl image file"
},
```

```
"mbl": {
    "flag": 1,
    "select": 1,
    "bin_path": "bootloader/bin/mbl_signed.bin",
    "flash_type": "qspi_nor",
    "description": "mbl image file"
},
"cp": {
    "flag": 1,
    "select": 1,
    "bin_path": "cp/bin/aicos_cp_flash_signed.bin",
    "flash_type": "qspi_nor",
    "description": "cp image file"
},
"cpram1": {
    "flag": 1,
    "select": 1,
    "bin_path": "cp/bin/aicos_cp_ram1_signed_compressed.bin",
    "flash_type": "qspi_nor",
    "description": "cp ram1 image file"
},
"cpram2": {
    "flag": 1,
    "select": 1,
    "bin_path": "cp/bin/aicos_cp_ram2_signed_compressed.bin",
    "flash_type": "qspi_nor",
    "description": "cp ram2 image file"
},
"bl": {
    "flag": 1,
    "select": 1,
    "bin_path": "bootloader/bin/bl_signed_compressed.bin",
    "flash_type": "qspi_nor",
    "description": "bl image file"
},
```

+21 common lines +10

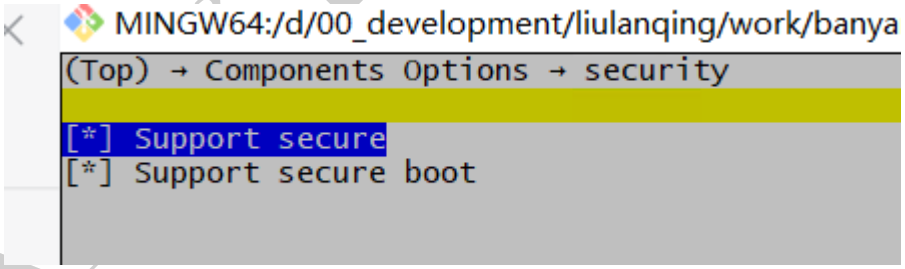
```
"md": {
    "flag": 1,
    "select": 1,
    "bin_path": "ap/bin/md_native.bin",
    "flash_type": "qspi_nor",
    "description": "misc data image file"
},
"ap": {
    "flag": 1,
    "select": 1,
    "bin_path": "ap/bin/aicos_ap.bin",
    "flash_type": "qspi_nor",
    "description": "ap basic image file"
},
"ap_extend": {
    "flag": 1,
    "select": 1,
    "bin_path": "ap/bin/aicos_ap_extend.bin",
    "flash_type": "qspi_nor",
    "description": "ap extend image file"
},
"calibration": {
    "backup": 1,
    "flash_type": "qspi_nor",
    "description": "calibration data partition"
},
"C": {
    "flash_type": "qspi_nor",
```

```
"md": {
    "flag": 1,
    "select": 1,
    "bin_path": "ap/bin/md_native.bin",
    "flash_type": "qspi_nor",
    "description": "misc data image file"
},
"ap": {
    "flag": 1,
    "select": 1,
    "bin_path": "ap/bin/aicos_ap_signed.bin",
    "flash_type": "qspi_nor",
    "description": "ap basic image file"
},
"ap_extend": {
    "flag": 1,
    "select": 1,
    "bin_path": "ap/bin/aicos_ap_extend_signed.bin",
    "flash_type": "qspi_nor",
    "description": "ap extend image file"
},
"calibration": {
    "backup": 1,
    "flash_type": "qspi_nor",
    "description": "calibration data partition"
},
"C": {
    "flash_type": "qspi_nor",
```

+25 common lines +10

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1.5 AP开启secure boot功能



二、芯片使能secure boot功能

在此目录下：

secure_boot.h aicos\components\include\secure

提供如下接口：

```
/**
*·aic_secboot_get_verified_hash()·--·get·verified·pubkey·hash,·this·function
*····do·all·signed·image·authorization·use·the·returned·hash.
*·@hash_ptr:·pointer·to·hash·memory.
*·@hash_len:·pointer·of·hash·size.
*·Return:·0·for·success,·other·for·error.
*/
int·aic_secboot_get_verified_hash(uint8_t·*hash_ptr,·uint32_t·*hash_len);

/**
*·aic_secboot_get_pubkey()·--·get·pubkey·from·image
*·@pp_pubkey:·pointer·to·pubkey·memory·pointer.
*·@pubkey_len:·pointer·of·pubkey·size.
*·Return:·0·for·success,·other·for·error.
*/
int·aic_secboot_get_pubkey(uint8_t·**pp_pubkey,·uint32_t·*pubkey_len);
/**
*·aic_secboot_get_image_pubkey_hash()·--·get·pubkey·from·specified·image
*·@image_name:·image·name.
*·@hash:·pointer·of·pubkey·hash.
*·Return:·0·for·success,·other·for·error.
*/
int·aic_secboot_get_image_pubkey_hash(const·char·*image_name,·uint8_t·*hash);
/**
*·aic_secboot_write_pubkey_hash()·--·write·pubkey·hash·to·efuse
*·@pubkey_hash:·pointer·of·pubkey·hash.
*·Return:·0·for·success,·other·for·error.
*/
int·aic_secboot_write_pubkey_hash(uint8_t·*pubkey_hash);
/**
*·aic_secboot_enable()·--·write·secureboot·enable·bit·to·efuse
*·Return:·0·for·success,·other·for·error.
*/
int·aic_secboot_enable(void);
/**
*·aic_rsa_verify()·--·do·rsa·verify
*·@hash:·The·buffer·holding·the·message·digest·or·raw·data.
*·@sign:·The·buffer·holding·the·signature..
*·@pub_key:·pointer·to·pubkey.
*·Return:·0·for·success,·other·for·error.
*/
int·aic_rsa_verify(uint8_t·*hash,·uint8_t·*sign,·struct·rsa_pubkey·*pub_key);
```

需要用上述接口先将pubkey的hash写入到efuse中，然后使能芯片的secure boot功能。通过aic_secboot_enable接口使能芯片secboot功能后，下一次重启就会走完整的验签流程。